

continuation

Continuation Document

Revision: 27 **Last modified:** 2026-06-01T09:36:01Z **Description:** Sacred invariant resumption document for Helix Cluster OS **Authority:** Constitution §12.10
Maintainer: Operator + AI loop

§0: How to use this document

Any CLI agent resuming work on this project MUST read this file first.

§1: Recently Completed Work (last 10 commits)

Commit	Message
4554e9a	Phase 0 enabling: integrate constitution upstream (§11.4.106 Docs Chain + workable-items DB), add docs_chain + helixqa submodules, purge stale test binaries, gitignore docs_chain runtime
279bce8	Bump security submodule -> fb1f3ce (Phase 8C e2ee interop + gputest spot-check/seal)
f73806e	Phase 8C gaps: model-integrity gate, attestation-gated admission, GPU class model, fiber stake admission, inference proxy (PCS-6/CLAUDE-1)
31c4490	Phase 8B gaps: GPU pool manager, cost broker, burst hysteresis, local GPU TCO, provider adapter (PCS-6/CLAUDE-1)
8b04418	Phase 8 gaps: chutes inference client+SSE+crypto seams, marketplace adapter+scorer, GPU dual-class reservation (PCS-6/CLAUDE-1)
ded59ff	Phase 7 gaps: hashslot CRC16, largest-subcluster voting, SLURM backfill, health startup tier, N+K admission, BOINC trust (PCS-6/CLAUDE-1)

Commit	Message
a31c0de	Phase 6 gaps: HLC, CRDTs+merkle anti-entropy, federation cell registry/FSM, SWIM phi-accrual (PCS-6/CLAUDE-1)
0d543b0	Phase 5 gaps: device tier classifier, tier+power scheduler plugin, trust registry, cloudspot preemption, inference router (PCS-6/CLAUDE-1)
7a58d0b	Phase 4 gaps: 26 chaos faults, SWIM SimTransport DST, Welch regression detector, device provisioner, wasm capability sandbox (PCS-6/CLAUDE-1)
2e70be1	Phase 3 gaps: classads scheduler plugin, S3 storage backend (SigV4), OTLP/HTTP trace exporter (PCS-6/CLAUDE-1)

§2: Environment Snapshot

Property	Value
Branch	main
Commit	4554e9a
Timestamp	2026-06-01T09:36:01Z

§3: Active Work

HXC	Title	Status
HXC-922	Node service, advisory locks, chaos tests, docs	✓ Done
HXC-921	Web UI, K8s/Helm, integration, E2E, benchmarks	✓ Done
HXC-920	Health, security, build, wireguard services	✓ Done
HXC-919	htmux, messaging, GPU, WASM, storage, build manifest	✓ Done
HXC-918	Wire scheduler/session gRPC to real backends, etcd discovery	✓ Done
HXC-916	Stub expansion (config, retry, ratelimit, validator, build cache)	✓ Done
HXC-914	LLM, policy, setup, distributed lock	✓ Done

HXC	Title	Status
HXC-913	Health, security, ClassAds parser	✓ Done
HXC-912	Gateway, helixd, helix-agent	✓ Done
HXC-911	Scheduler, Session gRPC services	✓ Done
HXC-910	Console, security, testing infra	✓ Done

§4: Next Planned Work

All MVP, Phase 2, Phase 3, and Phase 4 are **COMPLETE**. Potential future enhancements:

1. **Performance optimization** — Profile and optimize hot paths
2. **Multi-region support** — Cross-region cluster federation
3. **GPU scheduling v2** — Multi-GPU, fractional GPU allocation
4. **Web UI real-time** — WebSocket integration for live updates
5. **Operator pattern** — Kubernetes operator for cluster management

§5: Known Issues / Blockers

- None

§6: Quick Commands

```
# Run all tests (requires GOWORK due to workspace modules)
GOWORK=$(pwd)/go.work go test ./... -race -count=1
```

```
# Build all binaries
GOWORK=$(pwd)/go.work go build ./cmd/...
```

```
# Build web UI
cd web && npm run build
```

```
# Render Helm chart
helm template helix-cluster deploy/helm/
```

```
# Run benchmarks
go test -bench=. ./test/benchmark
```

```
# Run chaos tests
go test -race -count=1 ./test/chaos
```